DS1307 Real-Time Clock & I2C Subsystem

1. Overview of DS1307 RTC

The **DS1307** is a low-power, full binary-coded decimal (BCD) clock/calendar real-time clock IC manufactured by Analog Devices (formerly Maxim Integrated). It tracks seconds, minutes, hours, day of the week, date of the month, month, and year with leap-year compensation valid up to the year 2100.

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2. Technical Specifications

Parameter	Specification
Interface	I2C Serial Bus (Standard 100 kHz)
I2C Device Address	7-bit: `0x68` (STM32 HAL 8-bit shifted: `0xD0` Write, `0xD1` Read)
Operating Voltage (V_{CC})	4.5 V to 5.5 V (Nominal: 5.0 V)
Backup Battery (V_{BAT})	2.0 V to 3.5 V (Standard 3.0 V CR2032 Lithium Coin Cell)
Battery Backup Current	Less than 500 nA (enabling 10+ years of operation on a single coin cell)
Oscillator Frequency	32.768 kHz Quartz Crystal connected across `X1` and `X2`
Internal RAM	56 bytes of non-volatile battery-backed general purpose SRAM (`0x08` to `0x3F`)

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3. Register Map & Binary-Coded Decimal (BCD)

The DS1307 timekeeper registers occupy addresses `0x00` through `0x07`:

Address	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Function	Range
`0x00`	**CH**	10 Seconds	Seconds	Seconds	Seconds	Seconds	Seconds	Seconds	Seconds & Clock Halt	00–59
`0x01`	0	10 Minutes	Minutes	Minutes	Minutes	Minutes	Minutes	Minutes	Minutes	00–59
`0x02`	0	**12/24**	10 Hr/PM	10 Hr	Hours	Hours	Hours	Hours	Hours	00–23 (or 1–12)
`0x03`	0	0	0	0	0	Day	Day	Day	Day of Week	1–7
`0x04`	0	0	10 Date	Date	Date	Date	Date	Date	Date of Month	01–31
`0x05`	0	0	0	10 Month	Month	Month	Month	Month	Month	01–12
`0x06`	10 Year	Year	Year	Year	Year	Year	Year	Year	Year	00–99
`0x07`	OUT	0	0	SQWE	0	0	RS1	RS0	Control Register	-

4. Key Implementation Details in Firmware

4.1 BCD to Decimal Conversion

The DS1307 represents digits using separate 4-bit nibbles for tens and units.

- E.g. Minute 45 is stored as binary **`0100 0101b`** = **`0x45`**.
- Our driver provides rapid bit-shift conversion:

```
#define BCD_TO_DEC(val) (((val) >> 4) * 10 + ((val) & 0x0F))
#define DEC_TO_BCD(val) ((((val) / 10) << 4) | ((val) % 10))
```

4.2 Handling the Clock Halt (`CH`) Flag

- **Bit 7 of Register `0x00`** is the **Clock Halt (CH)** bit.
- When power is first applied with a fresh battery, the oscillator is stopped and **`CH = 1`**.
- In **`DS1307_Init()`**, the firmware reads register **`0x00`**, verifies if **`CH == 1`**, and if so, writes **`0`** back to bit 7 to start the 32.768 kHz oscillator.

4.3 Burst Reading Over I2C

Rather than requesting each register individually, **`DS1307_GetTime()`** sends the starting register address **`0x00`** and reads all 7 bytes continuously in one transaction:

```
HAL_I2C_Mem_Read(hi2c, DS1307_I2C_ADDR, DS1307_REG_SECONDS, 1, buf, 7, 100);
```

The internal register pointer of the DS1307 automatically auto-increments after each byte read.

4.4 Hardware Practicalities & Diagnostic Bus Scanner

- **Power Supply Requirement (5V vs 3.3V)**: The DS1307 has an internal power-fail comparator that trips if $V_{CC} < 1.25 \times V_{BAT} (\approx 3.75\text{ V} - 4.0\text{ V})$. Supplying 3.3V locks the chip into battery-backup write-protect mode and disables the I2C interface. Therefore, V_{CC} must connect to the **5V rail** of the Blue Pill.
- **V_{BAT} Floating Hazard**: If no coin cell is inserted, the V_{BAT} pin will float, which can prevent the I2C interface from acknowledging. To operate without a battery, the **`BAT`** header pin must be tied to **`GND`**.
- **Tiny RTC Trickle-Charge Circuit Warning**: Standard "Tiny RTC" breakout modules include diode **`D1`** and resistor **`R5`** forming a crude charging path meant for rechargeable **LIR2032** cells. If using ordinary disposable **CR2032** lithium coin cells, desoldering diode **`D1`** is required to prevent overcharging.
- **On-Screen I2C Diagnostic Scanner**: At boot, the firmware scans addresses **`0x01`** through **`0x7F`** and renders detected slaves on the OLED display (**`0x3C`** for OLED, **`0x68`** for DS1307, **`0x50`** for companion 24C32 EEPROM), immediately confirming physical connectivity.
- **Non-Blocking Software Clock Fallback**: If the DS1307 is disconnected or unpowered, the firmware falls back to **`HAL_GetTick()`** elapsed time interpolation, keeping the clock ticking with an alert icon (**`!`**) without halting medication reminders.

5. Potential Evaluator Questions & Answers

- **Q: Why does the DS1307 require a 32.768 kHz crystal?**
- *A: $32,768 = 2^{15}$. Passing this frequency through a 15-stage binary divider (flip-flop chain) yields an exact $1.000\,\text{Hz}$ (1-second) tick with minimal low-power CMOS logic.*
- **Q: Can we power the DS1307 with 3.3V?**
- *A: No. The DS1307 requires $4.5\text{ V} \text{ to } 5.5\text{ V}$. At 3.3V, its internal comparator detects an undervoltage condition and inhibits the I2C bus to protect registers.*
- **Q: What is the purpose of the CR2032 coin cell?**
- *A: When main power (V_{CC}) drops below $1.25 \times V_{BAT}$, the DS1307 internal power-switch circuit switches to battery power (V_{BAT}), maintaining the oscillator and 56 bytes of internal SRAM while consuming less than 500 nA.*
- **Q: Can the 3.3V STM32 communicate safely with the 5V DS1307?**
- *A: Yes.* **PB6** *and* **PB7** *on the STM32F103 are 5V-tolerant (FT) pins. When configured in open-drain mode with pull-up resistors tied to 3.3V, the high logic level safely conforms to both devices' I2C V_{IH} thresholds.*